[Tutorial] C Structure Exercise

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Exercise 1

- Code

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

struct Student {
    char name[20];
    int grade;
    int class;
    float average;
};

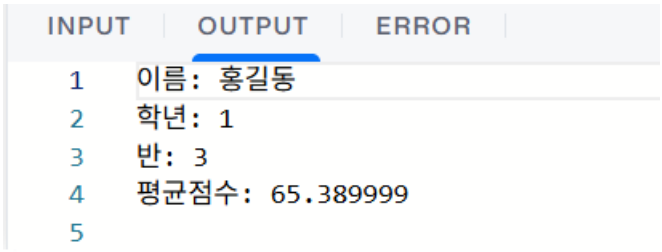
int main()
{
    struct Student *s1 =
malloc(sizeof(struct Student));
    strcpy(s1->name, "홍길동");
    s1->grade = 1;
    s1->class = 3;
    s1->average = 65.389999;

    printf("이름: %s\n", s1->name);
    printf("학년: %d\n", s1->grade);
    printf("반: %d\n", s1->class);
    printf("평균점수: %f\n", s1->average);

    free(s1);

    return 0;
}
```

- Output window



Exercise 2

- Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define _CRT_SECURE_NO_WARNINGS

typedef struct {
    char building_name[100];
    int room_number;
    char room_name[100];
} Handong;


int main()
{
    // Create structure type
(Handong)  variable  room1
    Handong room1;

    // Create structure type
(Handong)  variable  room2, room3
    Handong room2;
    Handong room3;


    // Create structure type
(Handong)  Pointer , roomPt
    Handong *room1Pt = &room1;

    // Assign room3 address to roomPt
    Handong *room2Pt = &room2;
    Handong *room3Pt = &room3;


    // Define structure
variable  memeber values: room1
    strcpy(room1.building_name,
"Newton");
    room1.room_number = 109;
    strcpy(room1.room_name, "iiLAB");

    // Define structure
variable  memeber values: room2
    strcpy(room2.building_name,
"Newton");
    room2.room_number = 118;
    strcpy(room2.room_name, "Control-
Lab");


    // Define structure
variable  memeber values: room3
    strcpy(room3.building_name,
"Newton");
    room3.room_number = 119;
    strcpy(room3.room_name, "SW-Lab");



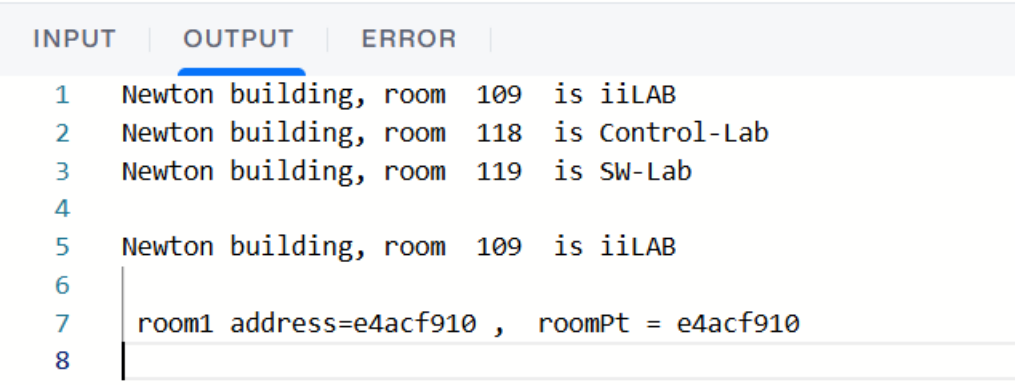
    // Print each member values  :
room1, room2, room3
    printf("%s building,
room  %d  is %s\n", room1.building_name,
room1.room_number, room1.room_name);
    printf("%s building,
room  %d  is %s\n", room2.building_name,
room2.room_number, room2.room_name);
    printf("%s building,
room  %d  is %s\n\n",
room3.building_name, room3.room_number,
room3.room_name);

    // Print each member values
by  pointer variable: room1Pt
    printf("%s building,
room  %d  is %s\n", room1Pt-
>building_name, room1Pt->room_number,
room1Pt->room_name);


    // Print address of   room1  and
value of room1Pt and compare.
    printf("\n room1
address=%x ,  roomPt = %x  \n", &room1,
room1Pt);

}
```

- Output window



Exercise 3

- Code

```
#include <stdio.h>
#include <math.h>
#include <stdlib.h>

typedef struct {
  int x;
  int y;
  int z;
} POSITION_TypeDef;

void addPos(POSITION_TypeDef pos1,
POSITION_TypeDef pos2,
POSITION_TypeDef *posOut);
void getDist(POSITION_TypeDef pos1,
POSITION_TypeDef pos2,
POSITION_TypeDef *posOut);
void printPos (POSITION_TypeDef Pos);

void main()
{

// Exercise 2
**************************************
*********
printf("\nExercise 2\n");
POSITION_TypeDef Pos[2];
POSITION_TypeDef PosOut;

Pos[0].x = 2; Pos[0].y = 2; Pos[0].z = 2;
Pos[1].x = 5; Pos[1].y = 5; Pos[1].z = 5;

addPos(Pos[0], Pos[1], &PosOut);
printPos(PosOut);
getDist(Pos[0], Pos[1], &PosOut);
printPos(PosOut);

system("pause");
}

void addPos(POSITION_TypeDef pos1,
POSITION_TypeDef pos2,
POSITION_TypeDef *posOut){
    posOut->x = pos1.x + pos2.x;
    posOut->y = pos1.y + pos2.y;
    posOut->z = pos1.z + pos2.z;
  }

  // getDist : get the distance between the
two points
  void getDist(POSITION_TypeDef pos1,
POSITION_TypeDef pos2,
POSITION_TypeDef *posOut){
    int xgap, ygap, zgap = 0;
    xgap = abs(pos1.x - pos2.x);
    ygap = abs(pos1.y - pos2.y);
    zgap = abs(pos1.z - pos2.z);

    posOut->x = xgap;
    posOut->y = ygap;
    posOut->z = zgap;
  }

  void printPos (POSITION_TypeDef Pos){
    printf("(x, y, z) = (%d, %d, %d)\n",
Pos.x, Pos.y, Pos.z);
  }
```

- Output window

